Name: _____

Making Some Ionic Compounds

A. Heat some magnesium ribbon over a portable Bunsen. Don't look at the flames. Wear safety glasses.

1. What colour was the material produced?

   _____

The material produced is Magnesium oxide.

When the Magnesium burns each magnesium atom loses 2 electrons and forms the ion

Mg^{+2}

The oxygen atoms in the air near the burning magnesium gain two electrons and form the ion

O^{-2}

These ions have different charges (positive and negative) so they will join together with _____ bonds.

2. Write the formula for the magnesium oxide.

3. Write the word equation then the formula equation showing magnesium burning in oxygen to produce magnesium oxide. Ensure it is balanced and don't forget to include state symbols.

B. Sprinkle some iron filings over the blue heating flame of the portable Bunsen. WEAR SAFETY GLASSES

1. What colour was the material produced?

   _____

The material produced is iron oxide.

When the iron burns each iron atom loses 3 electrons and forms the ion



The oxygen atoms in the air near the burning iron gain two electrons and form the ion



These ions have different charges (positive and negative) so the will bond together to form _____ bonds.

2. Write the formula for the iron oxide.

3. Write out the word equation then the formula equation showing iron burning in oxygen to produce iron oxide. Ensure it is balanced and don't forget to include state symbols.

C. Heat some fine copper wire over a portable Bunsen. Wear safety glasses.

1. What colour was the material produced?

2. What is the name of the material produced?

When the copper burns each copper atom loses 2 electrons and forms the ion

The oxygen atoms in the air near the burning copper gain two electrons and form the  ion

These ions have different charges (positive and negative) so the will bond together to form _____ bonds.

3. Write the formula for the copper oxide.

4. Write out the word equation then the formula equation that shows copper burning in oxygen to produce copper oxide. Ensure it is balanced and don’t forget to include state symbols.

Balancing Simple Chemical Equations.

Balance and write the names of the following formulas.

1. Al PO_4

   _____

2. Zn PO_4

   _____

3. Pb NO_3

   _____

4. Cu SO_4

   _____

5. Zn HCO_3

   _____

Complete the word equation then write the formula equation underneath

1. Hydrogen + Oxide --------->

2. Sodium + Hydroxide -------→

3. Copper + Fluoride --------->

4. Iron (III) + Carbonate --------->

5. Lead + Phosphate --------->

6. Tin + Nitrate --------->

7. Magnesium + Sulfate --------->

Balancing Chemical Equations

Balance the equations below:

1) ____ N_2 + ____ H_2 → ____ NH_3

2) ____ $KClO_3$ → ____ KCl + ____ O_2

3) ____ NaCl + ____ F_2 → ____ NaF + ____ Cl_2

4) ____ H_2 + ____ O_2 → ____ H_2O

5) ____ $Pb(OH)_2$ + ____ HCl → ____ H_2O + ____ $PbCl_2$

6) ____ $AlBr_3$ + ____ K_2SO_4 → ____ KBr + ____ $Al_2(SO_4)_3$

7) ____ CH_4 + ____ O_2 → ____ CO_2 + ____ H_2O

8) ____ C_3H_8 + ____ O_2 → ____ CO_2 + ____ H_2O

9) ____ C_8H_{18} + ____ O_2 → ____ CO_2 + ____ H_2O

10) ____ $FeCl_3$ + ____ NaOH → ____ $Fe(OH)_3$ + ____NaCl

11) ____ P + ____O_2 → ____P_2O_5

12) ____ Na + ____ H_2O → ____ NaOH + ____H_2

13) ____ Ag_2O → ____ Ag + ____O_2

14) ____ S_8 + ____O_2 → ____ SO_3

15) ____ CO_2 + ____ H_2O → ____ $C_6H_{12}O_6$ + ____O_2

16) ____ K + ____ MgBr → ____ KBr + ____ Mg

17) ____ HCl + ____ $CaCO_3$ → ____ $CaCl_2$ + ____H_2O + ____ CO_2

18) ____ HNO_3 + ____ $NaHCO_3$ → ____ $NaNO_3$ + ____ H_2O + ____ CO_2

19) ____ H_2O + ____ O_2 → ____ H_2O_2

CATIONS

1+ Charge		2+ Charge		3+ Charge	
H^{1+}	hydrogen	Mg^{2+}	magnesium	Al^{3+}	aluminum
Na^{1+}	sodium	Ca^{2+}	calcium	Fe^{3+}	iron(III)
K^{1+}	potassium	Ba^{2+}	barium	Cr^{3+}	chromium(III)*
Cu^{1+}	copper(I)	Zn^{2+}	zinc		
Ag^{1+}	silver	Cd^{2+}	cadmium		
NH_4^{1+}	ammonium	Hg^{2+}	mercury(II)		
Li^{1+}	lithium	Cu^{2+}	copper(II)		
		Pb^{2+}	lead(II)		
		Fe^{2+}	iron(II)		

ANIONS

1– Charge		2– Charge		3– Charge	
F^{1-}	fluoride	O^{2-}	oxide	PO_4^{3-}	phosphate
Cl^{1-}	chloride	S^{2-}	sulfide	P^{3-}	phosphide
Br^{1-}	bromide	SO_4^{2-}	sulfate	N^{3-}	nitride
I^{1-}	iodide	SO_3^{2-}	sulfite		
NO_3^{1-}	nitrate	CO_3^{2-}	carbonate		
NO_2^{1-}	nitrite				
OH^{1-}	hydroxide				
CN^{1-}	cyanide				
$CH_3CO_2^{1-}$	acetate				
HCO_3^{1-}	bicarbonate				
ClO^{1-}	hypochlorite				
SCN^{1-}	thiocyanate				
ClO_3^{1-}	chlorate				